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CORRESPONDENCE

Brain connectivity: complex, not chaotic

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Abstract
The term "connections" is a commonly used and convenient shorthand for describing the complex organization of the brain, but it can easily lead to an overemphasis on pairwise or point-to-point, source-target network connectivity. Anatomical studies make clear that there are other important features to consider such as divergence and collateralization (axons or bundles branching to multiple targets), convergence (multiple bundles from different sources converging on the same target), and scrambled topography along a trajectory. This short "Did You Know" communication elaborates on several of these features from the anatomical perspective, while inviting continued dialogue with the tractography community in addressing the shared goals of better understanding brain organization.

Introduction

Tractography and neuroanatomy share the goal of understanding the network interconnections of the central nervous system but approach the issue from different traditions using complementary methods. Methods differ in their strengths and limitations as well as in the scale at which they operate (respectively, voxels ~ 1mm^3 vs axons and bundles (< 1 µm–100 µm). An explicit goal of the 2024 TractAnat Retreat was to better articulate these different approaches and enhance communications across the combined communities. Five topics from this dialogue are developed here, with accompanying images (Figs. 1 and 2) selected to illustrate basic points from the anatomical perspective.

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Understanding brain connections

The term "connections" in neuroscience encompasses a range of meanings, including structural, functional, and effective connectivity, or something more abstract, as in a graph theoretical context (e.g. Fraskowitz et al., 2022; Seguin et al. 2023). In this section, we focus on anatomical aspects of connections, rather than on "connectivity" in the broader or computational sense of networks and systems. In a simple anatomical definition, "connections" can be taken to refer to multidimensional and likely dynamic whole brain interactions under activity-driven or other conditions (Rockland 2015).

Connections originate from and terminate on neurons. Anatomical approaches have tended to give a prominent place to the pre- and postsynaptic gray matter components, with axon trajectories and white matter organization receiving less attention ("conduction" component, as distinguished by Innocenti 1995; Innocenti and Caminiti 2017, among others). This is largely due to the constraints of tracer injections in experimental animal models and limitations of what can be visualized at the microscopic level over long distances. This situation is rapidly improving with better anatomical techniques, which allow for 3-dimensional whole neuron and whole brain visualization.

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As a term, “connections” is a convenient shorthand that successfully evokes the multiple aspects of brain organization across spatial scales. It also inevitably gives rise to over-simplification, and one can inadvertently forget that “area A projects to area B” by axons not arrows. Here, we summarize five examples chosen to highlight features of brain anatomy that may not be broadly recognized but have relevance to tractography with regard to technical developments, novel experimental design, and richer and more precise interpretation of results.

Bundles as well as axons have complex white matter trajectories, including right angle turns

The concept of right angles can be applied to the geometric organization of white matter (WM) in two distinct ways. **First**, as known already from the myelin-based maps of Dejerine, Golgi preparations of Cajal, and gross dissections by the Klingler method, bundle trajectories can incorporate 90° bends (e.g., Meyer’s loop in Fig. 1a). The ansa lenticularis sharply loops ventral to the globus pallidus before continuing medially and posterior. The superior sagittal stratum has an L-shaped medio-lateral elbow bend in its ventral component, as this passes ventral to the lateral ventricle (Fig. 1b, c). Individual axons, although traveling within the perimeter of the bundle (and observing the elbow-bend), have an anterior–posterior trajectory (Fig. 1d). **Second**, right-angle branches frequently occur along individual axons or small groups of axons. Abrupt changes in trajectory can come from the main axonal trajectory or from collaterals (Fig. 1e). An axonal right-angle configuration can also be observed in rodent brains when callosal neurons enter the cortex (e.g. Zhou et al. 2013; Rayêe et al. 2021).

Axons have collateral branches to multiple targets

Connections between brain regions are often assumed to be composed of axons that exclusively connect one region to another (“pairwise” or point-to-point connections). A primary terminal target from a single source often does originate from specialized and distinct neuronal subpopulations (“dedicated” 1:1 source to target, projections). As increasingly documented, however, there are also “broadcast” projections, where one neuron in the source area projects to multiple spatially separate targets via branched collateral projections (Fig. 1e; and see Parent et al. 2000; Han et al. 2018). Although less numerous than in the developing brain, “broadcast” projections are present throughout the adult brain. A target structure can receive, in variable proportions, both dedicated (unbranched) and broadcasting (branched) projections from different neurons in the same originating source. Although collateral branches may not form distinct bundles, they can impact the putative fiber orientations of a voxel measured by diffusion MRI, and result in false positive streamlines.

As a rule, cortical pyramidal neurons with extrinsic connections (cortical or subcortical) typically in addition have extensive intrinsic collaterals within a 2-3 mm radius of the parent soma. Collateralized (“broadcast”) projections are characteristic for cortical layer 5b corticothalamic projecting neurons, but not the layer 6 corticothalamic (Rockland 2019). Corticostriatal projections can collateralize to both the ipsi- and contralateral striatum and possibly the contralateral cortex (Borra et al. 2022). An important point is that the set of structures targeted by the collateral branches is not fixed. As an example, the functionally significant hyperdirect pathway from the motor cortex derives from collaterals of corticofugal axons that also innervate the red nucleus *and/or* zona incerta (Coudé et al. 2018). In the early cortical visual system, large Meynert cells project to extrastriate visual areas *and/or* the superior colliculus (Vogt Weisenhorn et al. 1995).

Axons may not always be topographically organized within a bundle

“Topographic organization” refers to the reproducibly ordered spatial arrangement of neurons in the gray matter and the preservation of this arrangement by their outgoing axons as these travel through the white matter. Topography commonly refers to somatotopic representations (the classical mapping of the “homunculus”), but also applies to preserved positions in anterior–posterior or medio-lateral axes, an example being the location of outgoing fibers from the anterior or posterior hippocampus in the fornix (Christiansen et al. 2016). Continuing investigations, especially in animal models, are clarifying that topographic maps at a cortical origin are not necessarily strictly maintained at all locations along the outgoing axon bundles. Taking the corticospinal tract as an example, descending axons from somatotopically adjacent cortical gray matter maps intermingle in the pons and spinal cord, without apparent reference to initial somatotopic adjacency as mapped by tracer injections in macaque monkeys (Fig. 1f; and Morecraft et al. 2017, 2022; Lemon and Morecraft 2023).

The question of physical adjacency and reorganization is actively addressed by injections of double, colorimetrically distinguished tracers in macaque monkeys, with some emphasis on the internal capsule (Lehman et al. 2011; Safadi et al. 2018). Fewer studies are available for fine-grained organization of long-range corticocortical projections, where there are problems of access (for injections) and coverage (sulcal vs. gyral locations). A study of the macaque superior longitudinal fasciculus determined that axons exiting from a parietal brain area are spatially dispersed within the tract

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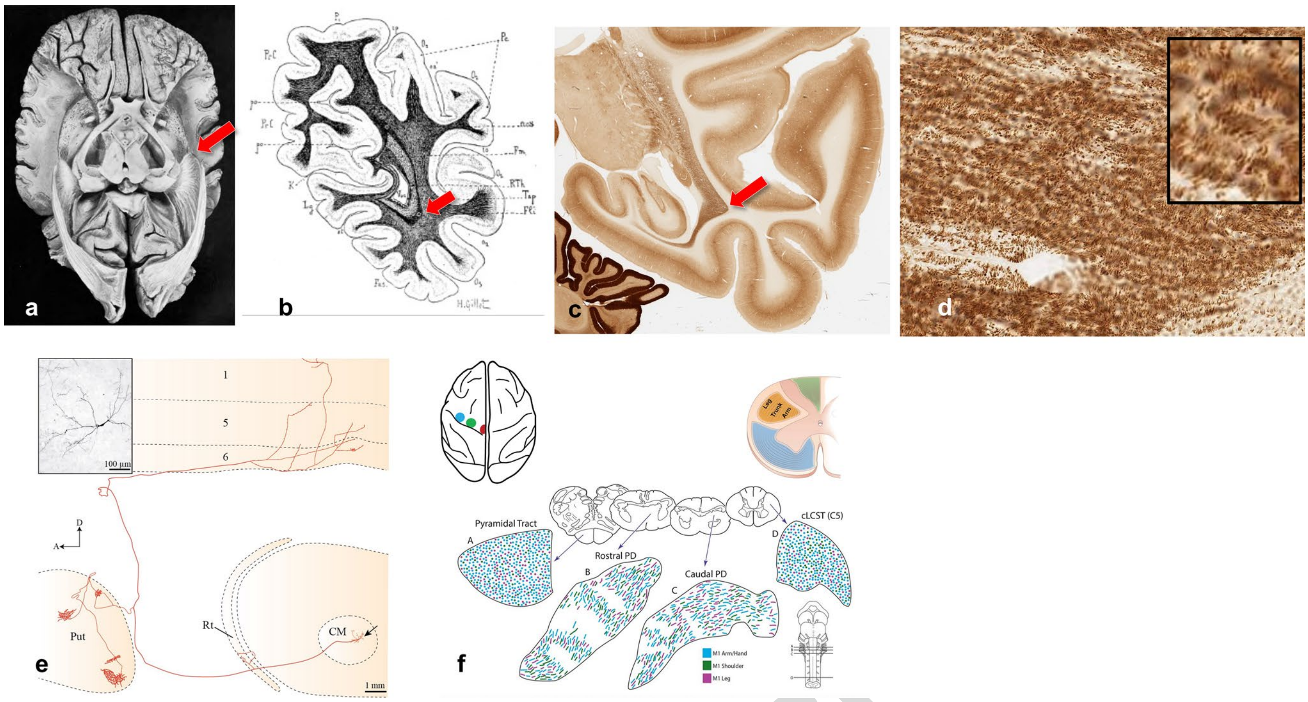

Fig. 1 Top row ("right-angle turns"). **a** Ventral view of the human brain, where a gross dissection (Klingler method) reveals the optic radiations coursing posterior from the thalamus into the occipital lobe, after making an anterior deviation ("Meyer's loop," arrow) (Ludwig and Klingler 1956). **b** Coronal histology section through the human occipital lobe, where a myelin stain reveals the optic radiations and the characteristic (lateral to medial) right angle, elbow bend at the ventral sector (arrow) (Dejerine and Dejerine-Klumpke 1895). Compare with **c** photomicrograph of a coronal section through the macaque monkey occipital lobe, where postmortem processing for parvalbumin (PV + thalamo-cortical projections) reveals a homologous configuration of the optic radiations, including the ventrally situated right-angle bend (arrow). The bend persists through several millimeters, anterior–posterior. **d** Higher magnification of the densely packed PV + thalamo-cortical axons in the optic radiation. These travel predominantly anterior–posterior within the tract and, in coronal sections, present as short segments, more obvious in the higher magnification box at upper right. From MacBrain Resource Center (Yale University), Brain 61 Sect. 42 (parvalbumin). Lower row ("collaterals"). **e** A sagittal plane reconstruction of a thalamostriatal neuron (arrow) in the centromedian nucleus (CM) of a cynomolgus monkey (photomicrograph of the soma is at upper left). There are a few collaterals to the reticular nucleus (Rt) and abundant arborizations, via collaterals, to the putamen (Put) and the deep layers of the primary motor cortex. Note the right-angle branch off the main axon to the putamen and the right-angle bend in the main axon as this approaches the terminal field in the motor cortex (Parent and Parent 2006, and Parent and Parent, Chapter 2 in Rockland 2016). ("topography"). **f** In the distal portion of the corticospinal tract (in the pons and spinal cord), the topographic organization according to body part is scrambled. (PD = pyramidal decussation, cLCST = caudal lateral corticospinal tract) (modified with permission, from Fig. 7, Lemon and Morecraft 2023)

itself, as assessed about 1.0 cm from the cortical injection site (Howells et al. 2020).

Overall, these results raise a cautionary note about assuming that intra-bundle axons en route to a common target remain aligned in the same parallel order. High-resolution approaches such as serial reconstructions reinforce that small groups of axons can braid together (Fig. 2a; Rockland 2018b; Andersson et al. 2020); and ongoing work is further delineating pathway specific patterns of tortuosity (e.g., Kjer et al. 2025).

Axonal density is a mis-interpreted term

Interpreting "fibre density" from diffusion MRI remains fraught with challenges due to both underlying anatomical complexity and methodological limitations. Two main analytical approaches attempt to approximate axonal architecture. Tractography produces the commonly used streamline count—the number of reconstructed streamlines passing through a voxel. Crucially, this reflects algorithmic output, not actual axon count. Yet the term "fibre density" is often misapplied here. We caution against this (Jones et al. 2013), as it lends biological specificity to a metric highly sensitive to acquisition settings (e.g., resolution, SNR, diffusion directions) and, more significantly, to processing choices like seeding density, tractography algorithm, and ROI definitions (Jones 2010; Yeh et al. 2021), making streamline count an unreliable proxy for axonal density. Biophysical modelling assigns parts of the diffusion MRI signal to intra- and extra-axonal compartments. Signal attributed to "restricted diffusion" is often interpreted as the intra-axonal volume fraction—and, problematically, as axonal density. However,

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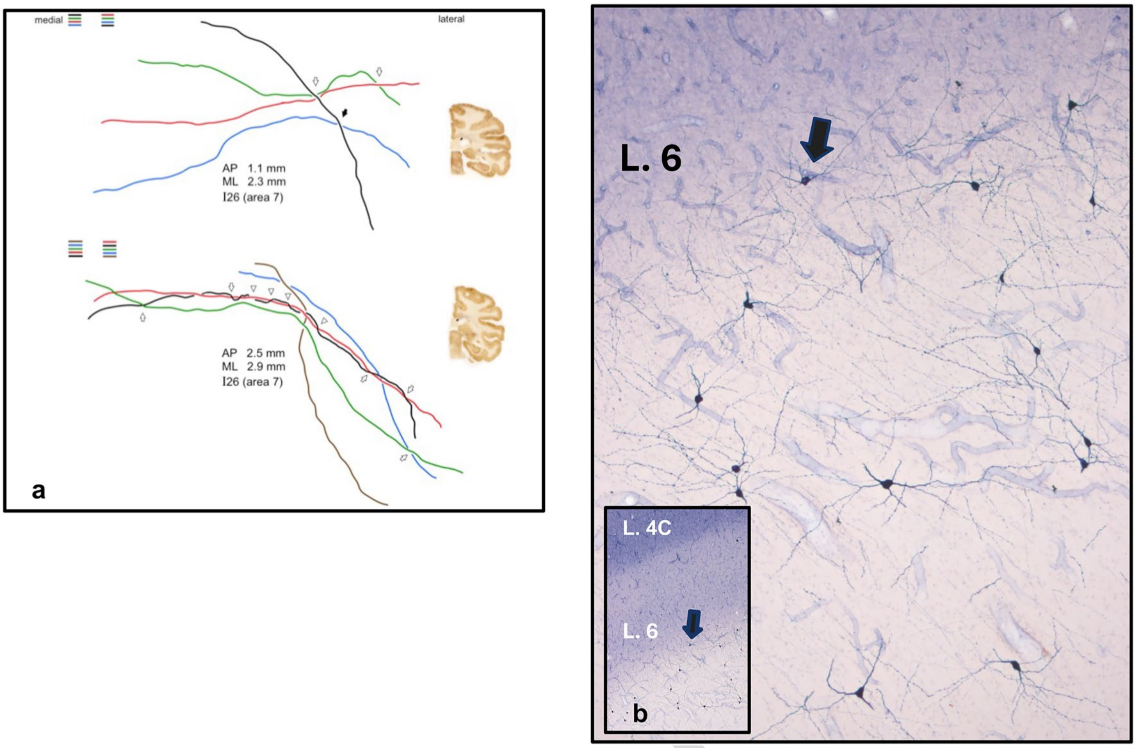

Fig. 2 ("intertwined callosal axons"). **a** Two groups of four and five axons each are anterogradely labeled by an in vivo injection in macaque monkey parietal cortex, and serially reconstructed (after perfusion and histological processing) for several millimeters (as shown). There are multiple intertwining events (solid arrow) and crossings (hollow arrows). By color coding the individual axons, it becomes apparent that the order is shuffled even over these relatively short distances. The coronal sections at right indicate the level where these axons are crossing in the corpus callosum (from Fig. 9, Rockland 2018a). ("WM neurons"). **b** The small subpopulation of white matter neurons positive for NADPH diaphorase is shown in a coronal section through the macaque visual cortex. This includes both superficial and deeper white matter. Layer 4C and layer 6 can be seen in the lower magnification view (box at lower left), where the arrow points to the same neuron at low and high magnification

volume fraction reflects space occupied, not axon count. A few large axons may produce the same signal fraction as many small ones, despite markedly different true fibre densities. Greater precision in terminology is essential to avoid conflating distinct biological properties and to bridge neuroimaging with neuroanatomy in a meaningful way.

There are neurons in the white matter

A common assumption is that there is a relatively sharp boundary between the cell dense gray matter (GM) and axon-rich white matter (WM). In point of fact, there are abundant myelinated axons in the cortical GM and, less widely recognized, abundant neurons in the WM (AKA "interstitial neurons," Fig. 2b). WM neurons (WMNs) were already noted by early investigators, who distinguished a sparse population of neurons in the superficial WM and sparser yet in the deeper cortical WM. More recent studies establish that WMNs are phylogenetically conserved (cf. references in Mortazavi et al. 2016; Fischer and Kukley 2024), and comprise about 50% excitatory and 50% mixed inhibitory types (García-Marín et al. 2010). Area- and species-specific differential densities are reported (in humans: García-Marín et al. 2010), and densities are reported to vary in several pathological conditions. The population of WMNs is known to be more abundant during development. At least some WMNs project into the gray matter, both overlying and distant (in vitro, in rat: Clancy et al. 2001; in vivo in macaque: Tomioka and Rockland 2007). In the WM, dendrites often, but not always, have a pronounced vertical orientation, parallel to the surrounding fiber orientation. Synaptic contacts (origin unknown) have been demonstrated (García-Marín et al. 2010), but much less is known about the WM-WM or WM-GM axonal arborizations.

Conclusion

In this brief overview, we presented five examples of widely misinterpreted or overlooked neuroanatomical facts. One conclusion could be that the search for general principles too often leads to inappropriate or at least premature and

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detrimental simplification. This results in an unfortunate amalgamating of the wonderful and necessary complexity of brain organization.

Brain organization spans multiple spatial scales, from micro (cellular and subcellular) to macro (white matter bundles, cortical areas and subcortical nuclei). To some extent, the full complexity is most evident at the micro level where specific details can be related to questions of cell types, input and output mappings at the synaptic level, connections that are dedicated (one source: one target) or broadcast (collaterals from one source: multiple targets), fasciculation and adjacency along a trajectory, heterogeneity and variability. Ultimately, however, the details will need to be functionally integrated with large scale organization, as the brain itself apparently does. The inherently limited microscopic resolution of diffusion MRI data, regardless of the species studied, constrains tractography to a simplified and often coarse representation of this rich complexity.

To a first view, the brain may look chaotic, but this perception-misperception is telling us that we need to consider complexity on its own terms. None of this is negative. In fact, progress can already be seen in technical advances that enable micro-scale visualization in a 3D whole brain framework. An additional approach, to which we hope to have contributed, is increased acknowledgement of the crucial need for multi-disciplinary coming together and conversation.

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Declarations

Conflict of interest The authors declare no competing interests.

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